

Signal Processing for Radio Astronomy Applications

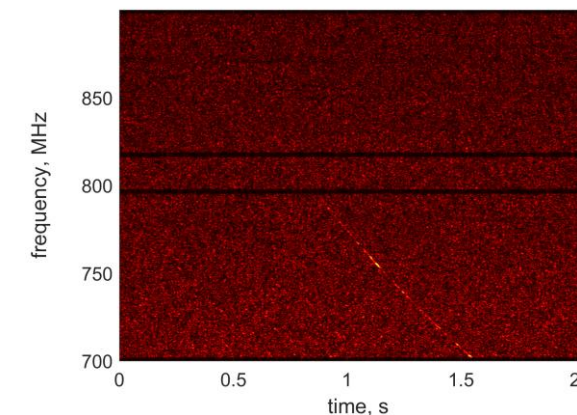
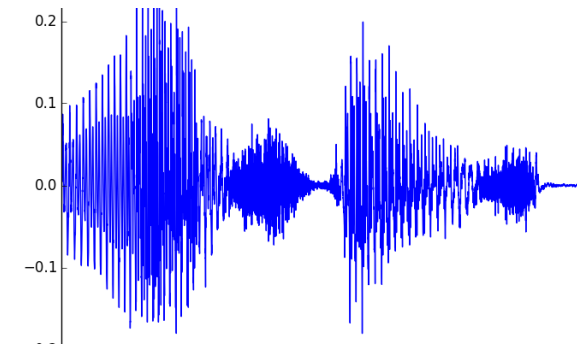


Signals

Signals = functions in 1D, 2D, 3D, etc.

Examples: audio signal, images, video signal

Dependence on time and frequency (or space and spatial frequencies)



Systems

Cellular antenna

Camera

Tomographer (CT, MRI, etc.)

Optical telescope

Radio telescope

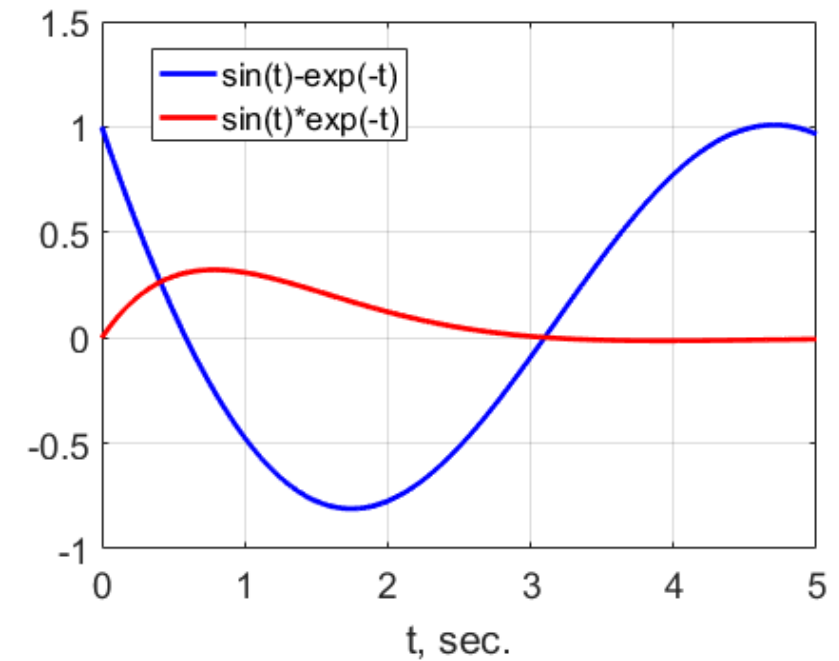
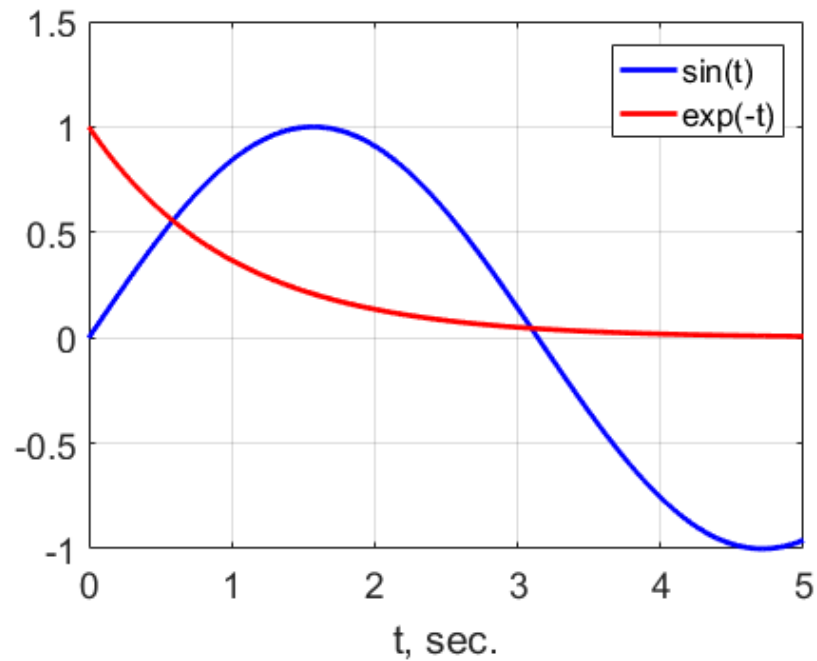
Radar

Any other receiver or antenna

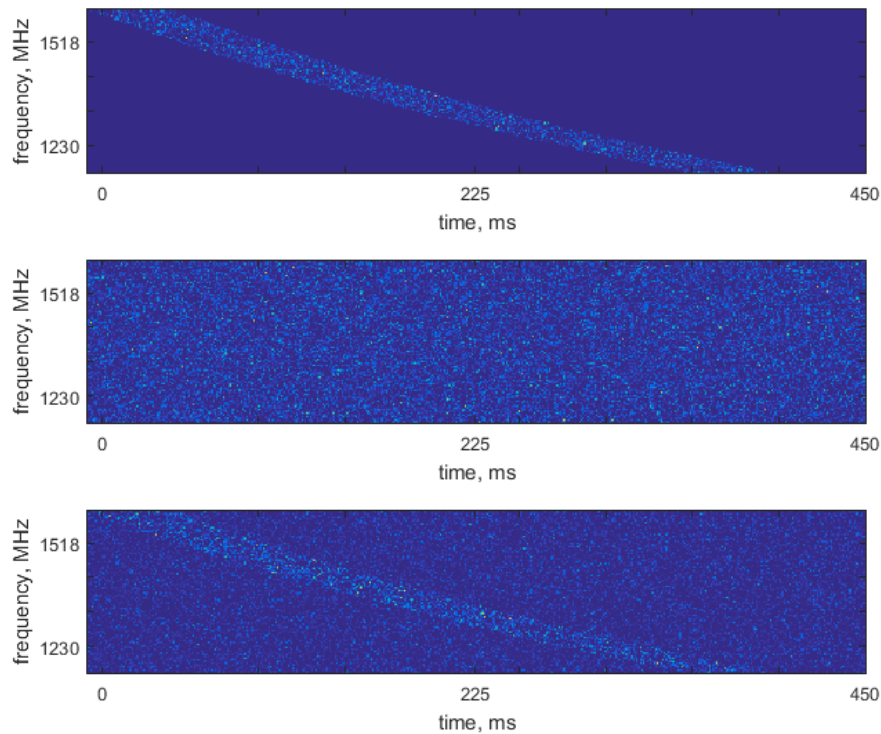


Operations on Signals

Adding / Subtracting / Multiplying



Signal processing for radio astronomy



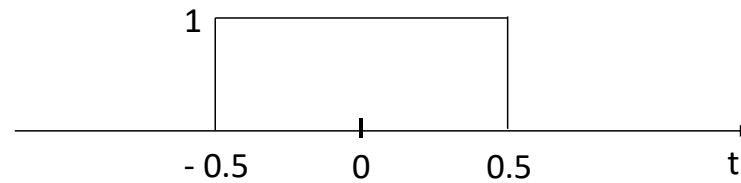
Adding signals:
radio astronomy

Simulated Lorimer
burst

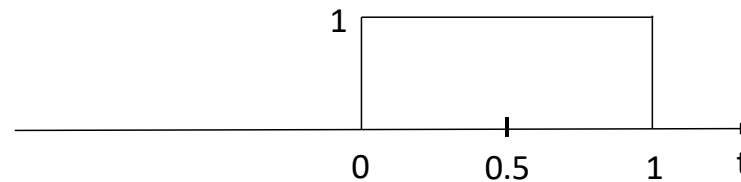
$DM = 375$

$S/N = 25$

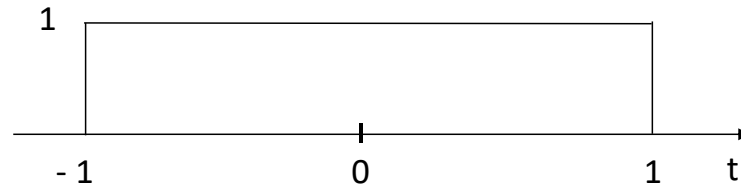
Time delay and scaling



$$\Pi(t) = \begin{cases} 1, & \text{if } |t| \leq 0.5 \\ 0, & \text{if } |t| > 0.5 \end{cases}$$



$$\Pi(t - 0.5)$$



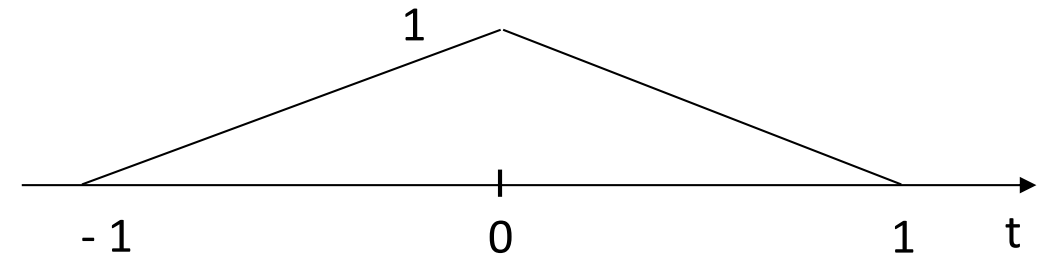
$$\Pi(0.5t)$$

Correlation of signals

$g(t)$ and $h(t)$ are two real valued signals
Correlation of $g(t)$ and $h(t)$ is defined as

$$R(t) = g(t) * h(t) = \int_{-\infty}^{+\infty} g(\tau)h(t + \tau)d\tau$$

Example: $\Pi(t) * \Pi(t)$



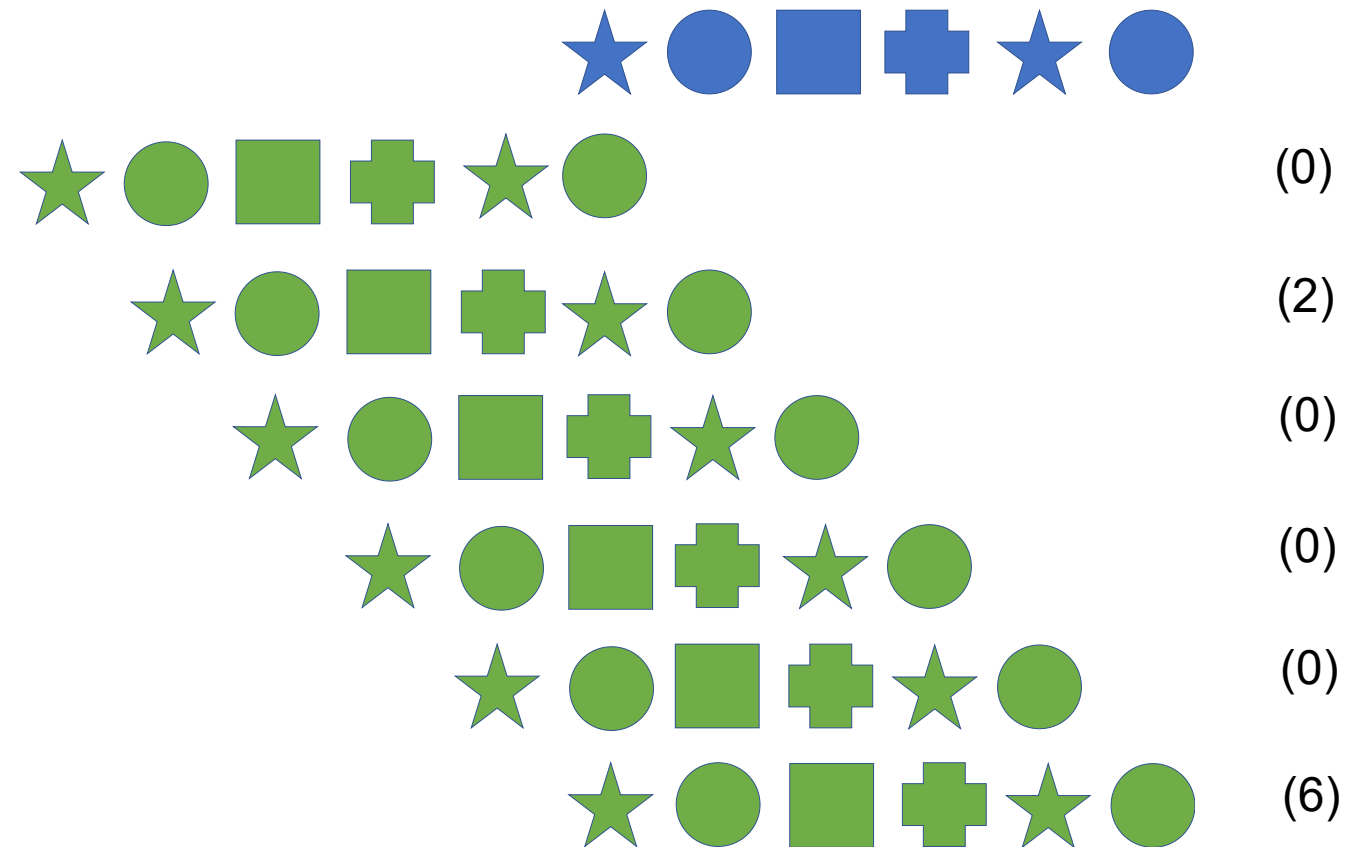
Why correlation?

It's like matching patterns

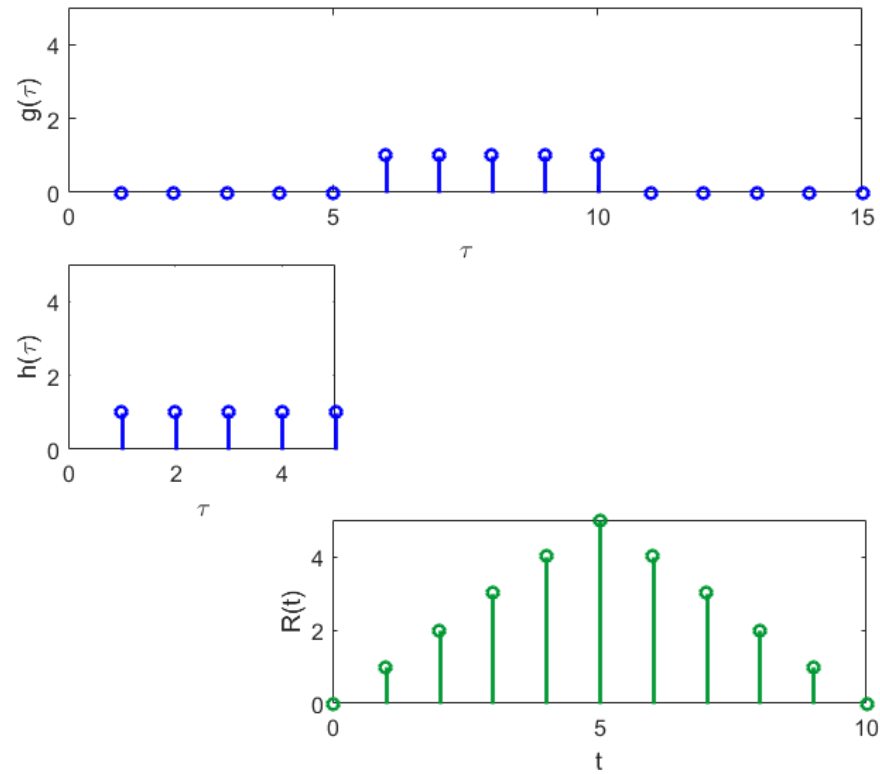
Operators: shift, logical
comparison

Assign one if two objects match
(zero, otherwise), add all ones,
find the maximum score. This is
the best match.

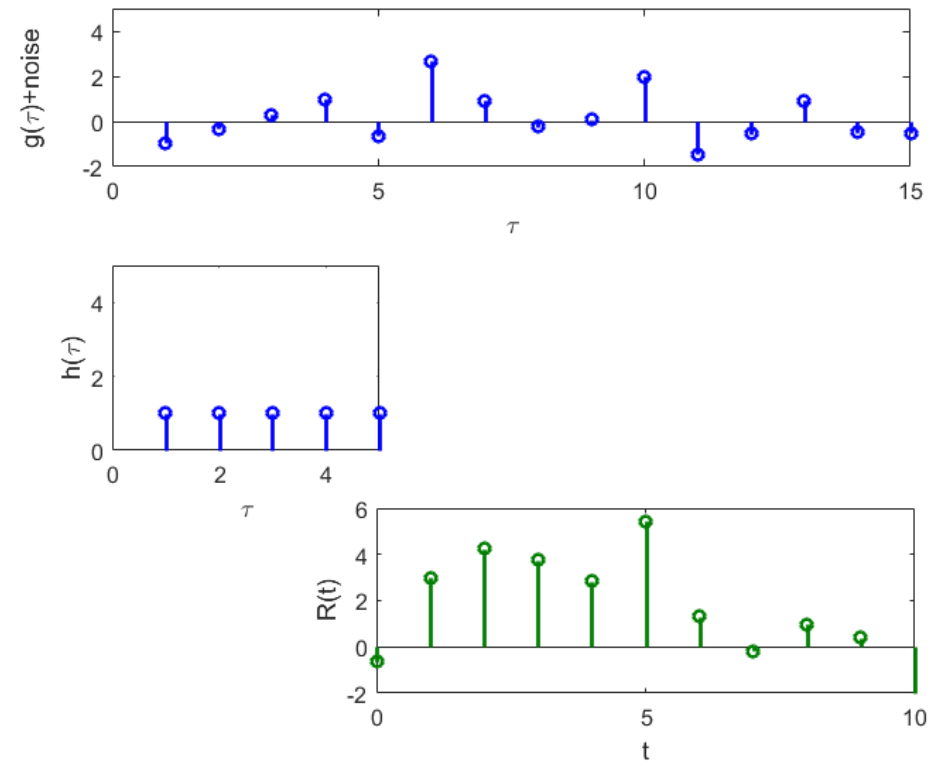
The same idea is used to
search for signals.



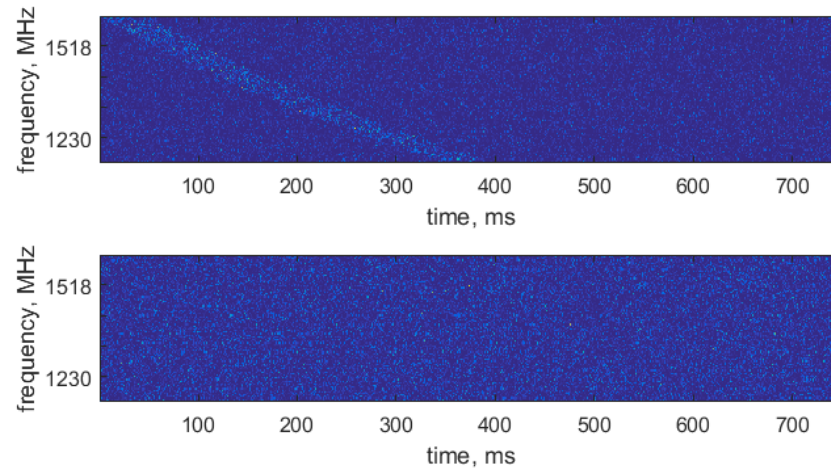
Illustration



Noisy signal



Matching signals in 2D



Signal + Noise	Noise only
1984.6	726.8
1013.0	740.3
896.8	755.8

